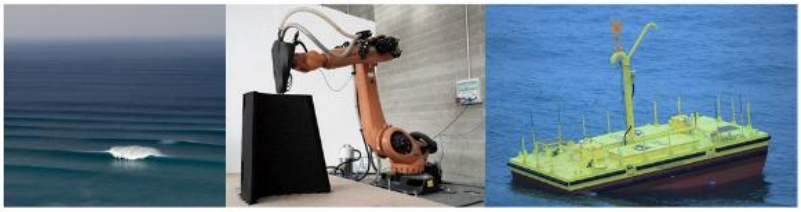
WEC-Pro Project
Work Package 1 Report



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Project title:

Progressing Wave Energy Conversion via Large-Scale Additive Manufacturing

Work Package 1:

Wave energy converter design for power absorption

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2. Executive summary

The WEC-Pro (Progressing Wave Energy Conversion via Large-Scale Additive Manufacturing) project addresses the high R&D costs and low Technology Readiness Level (TRL) currently hindering the commercialisation of wave energy technologies. Work Package 1 (WP1) focuses on the preliminary design and hydrodynamic optimisation of Wave Energy Converters (WECs). By integrating Large-Scale Additive Manufacturing (LSAM) constraints with advanced numerical modelling, WP1 aims to identify high-performance, cost-effective, and physically printable WEC geometries.

A central innovation of WP1 is the transition from conventional, simplified hull shapes to a cubic B-spline parametrisation. This approach unlocks a higher degree of geometric freedom, allowing for complex, organic hull designs that can be finely tuned to specific local wave climates. This methodology was successfully applied to two distinct WEC archetypes:

- PeWEC (Pendulum WEC) Optimisation: Utilising a frequency-domain hydrodynamic solver coupled with a Genetic Algorithm, thousands of configurations were evaluated to minimise the Cost of Energy (CoP). The novel spline-based hull significantly outperformed traditional arc-based designs. It achieved a higher Annual Energy Production (up to 80 MWh compared to the 60 MWh limit of the arc design) due to a more slender profile that reduced pitch radiation damping and enhanced the device's resonant response.
- Point Absorber (PA) Development: An automated Monte Carlo simulation framework was developed to generate 50,000 stochastic, spline-based hull profiles tailored to the wave climate at the NOEL laboratory. These profiles were subjected to rigorous LSAM validation filters to ensure manufacturability, such as strict monotonicity checks and a maximum overhang angle of 60°. Ultimately, 1,130 physically printable geometries were successfully isolated for deep hydrodynamic evaluation, aimed at maximising the Index of Performance (energy yield normalised against the wet surface area).

The numerical frameworks developed in WP1 successfully bridge the gap between theoretical hydrodynamic efficiency and practical manufacturing viability. This work establishes a robust foundation for the project's subsequent structural analysis (WP2) and experimental validation (WP3) phases. Key deliverables

stemming from this work package include open-access simulation scripts on Zotero and two peer-reviewed conference publications (EWTEC 2025 and OMAE 2026).

3. Acknowledgements

This project has received funding from the European Research Executive Agency (REA) under the European Union's Horizon Europe research and innovation programme under the Marie Skłodowska-Curie grant agreement No. 101206509.

4. Disclaimer

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6. Introduction

Context and Motivation

The global demand for green energy production has highlighted the immense, yet largely untapped, potential of sea waves as a renewable resource. Wave Energy Converters (WECs) are the primary technology required to harness this power to generate electricity. However, the sector currently faces significant hurdles, characterized by a low Technology Readiness Level (TRL) and high development and scaling costs, which have prevented wave energy from becoming a mainstream energy source.

A critical barrier to the advancement of WECs is the high cost associated with Research and Development (R&D), particularly during the prototyping and testing phases. Conventional manufacturing processes for offshore structures, such as mold preparation, welding, and assembly, are resource-intensive and demand high capital investment. These traditional methods limit the ability to rapidly iterate and test designs, which is essential for avoiding operational failures in later project stages. To rapidly increase the TRL of WECs and reduce R&D costs, there is an urgent need for strategies that enable efficient testing and production of WEC technology.

In this context, the WEC-Pro (Progressing Wave Energy Conversion via Large-Scale Additive Manufacturing) project proposes a paradigm shift by introducing cutting-edge Large-Scale Additive Manufacturing (LSAM) into the WEC development cycle. The primary drive of WEC-Pro is to empower the R&D of WECs to reach industrialization via LSAM, enabling the creation of optimized, high-performance devices that are both cost-effective and reliable.

This project is a 2-year Marie Sklodowska-Curie Actions (MSCA) fellowship carried out at Politecnico di Torino (POLITO) at the Marine Offshore Renewable Energy Lab (MORE), with strategic secondments at Caracol AM, a leading company in the LSAM sector.

Focus of Work Package 1 (WP1)

This report details the activities and outcomes of Work Package 1 (WP1): Wave energy converter design for power absorption. While the broader project addresses

structural analysis and physical testing, WP1 is dedicated to the preliminary design and hydrodynamic optimization phase.

By coupling time-domain models with genetic algorithms and LSAM constraints, WP1 aims to identify optimal configurations from a vast number of designs, ensuring that the resulting geometries are not only hydrodynamically efficient but also viable for printing. This establishes the foundation for the subsequent structural analysis (WP2) and experimental validation (WP3) phases of the WEC-Pro project.

7. Objectives of WP1

The primary objective of Work Package 1 is to further advance existing numerical tools to develop more performing WEC geometries. This involves adapting nonlinear time-domain models and genetic algorithms to optimize designs for maximum power absorption while adhering to specific manufacturing constraints.

The adoption of LSAM is central to this approach, as it allows for the manufacturing of more complex geometries at competitive costs compared to traditional methods. Consequently, this work investigates more flexible hull shapes to fully exploit this manufacturing freedom. These complex geometries are defined and investigated using spline shapes, enabling a broader exploration of the design space.

This methodology is applied to two specific case studies. The first is the PeWEC, an inertial WEC device. The second case study selected is a Point Absorber, which is an oscillating type of WEC.

While the initial project proposal considered the PivWEC, which is also an oscillating type of WEC, the Point Absorber was ultimately selected for this work package. This decision is justified by the fact that the MORE Lab has extended experience with this specific Point Absorber technology (WEPA WEC also deployed at sea for full scale testing), which has reached a more advanced level of commercial development compared to the PivWEC. Furthermore, the Point Absorber configuration was determined to be more reliable to evaluated during the experimental testing phases (WP3) planned for the project, ensuring a more robust validation of the numerical tools developed.

8. Numerical modelling and development of PeWEC

This section details the numerical framework established in WP1 to simulate the system dynamics, estimate power production, and evaluate the techno-economic performance of the PeWEC. The approach integrates a frequency-domain hydrodynamic solver with a genetic algorithm (GA) to enable the rapid evaluation of thousands of design configurations.

To minimize expensive marine maintenance, the Pendulum Wave Energy Converter (PeWEC)-investigated in this study-utilizes a fully enclosed Power Take-Off (PTO). Typical of inertial WECs, this configuration ensures that sensitive electromechanical components are shielded from the corrosive marine environment.

The design process for WECs is inherently a complex task. Historically, common WEC design approaches have been sequential and, therefore, suboptimal, where the hull geometry is often defined first based on hydrodynamic principles, followed by the selection of the PTO. However, an effective design requires a holistic approach based on representative wave-to-wire mathematical models. This aspect is critical because the implementation of an energy-maximizing control strategy introduces strong nonlinearities and fundamentally alters the dynamic response of the device. As a result, energy absorbed-wise optimal solutions for uncontrolled versus controlled conditions can differ significantly. Furthermore, the hydrodynamic behavior depends mainly on shape and dimensions, linked to volume and mass, which are the primary drivers of capital cost (CapEx). Navigating this trade-off between energy production and cost requires Multidisciplinary Design Optimization (MDO). Yet, a practical implementation of an all-encompassing MDO is challenging due to the vast, multi-dimensional search space.

In this context, both the cost function and the physical constraints often behave as grey or black boxes, since the system cannot be described by a single mathematical formula or a simple combination of equations. Consequently, Genetic Algorithms (GAs) have become a common choice. As metaheuristic optimization codes based on evolutionary theory, GAs rely on the survival-of-the-fittest principle. They are particularly well-suited for high-dimensionality problems where the mathematical structure of the objective function is unknown or non-differentiable. Thanks to their stochastic nature, GAs favor the identification of global optima, reducing the risk of

converging to local minima inherent in deterministic methods. To date, the application of Genetic Algorithms within the wave energy sector has been relatively limited. While other metaheuristic approaches such as particle swarm optimization or differential evolution exist in the general marine engineering field, their application in wave energy has often been limited to array layout optimization. Device-specific optimization has seen limited but notable efforts, often optimizing basic parameters like the radius and draft of an archetypal cylinder or fixed-axis devices of conventional shapes. More recently, efforts have been made to break away from simple primitives, for instance by attempting to generalize geometry using B-splines.

This report presents a recent study extending the existing Genetic Algorithm-based MDO model of the PeWEC, developed at the MORE research group of Politecnico di Torino. The primary advancement lies in the integration of a spline-based parametrization for the WEC hull. Previously, the optimization model relied on a simplified hull geometry defined by basic arcs. By adopting a spline-based approach, this study allows for a higher degree of geometric freedom, enabling the exploration of complex shapes that can significantly outperform standard geometries in hydrodynamic efficiency.

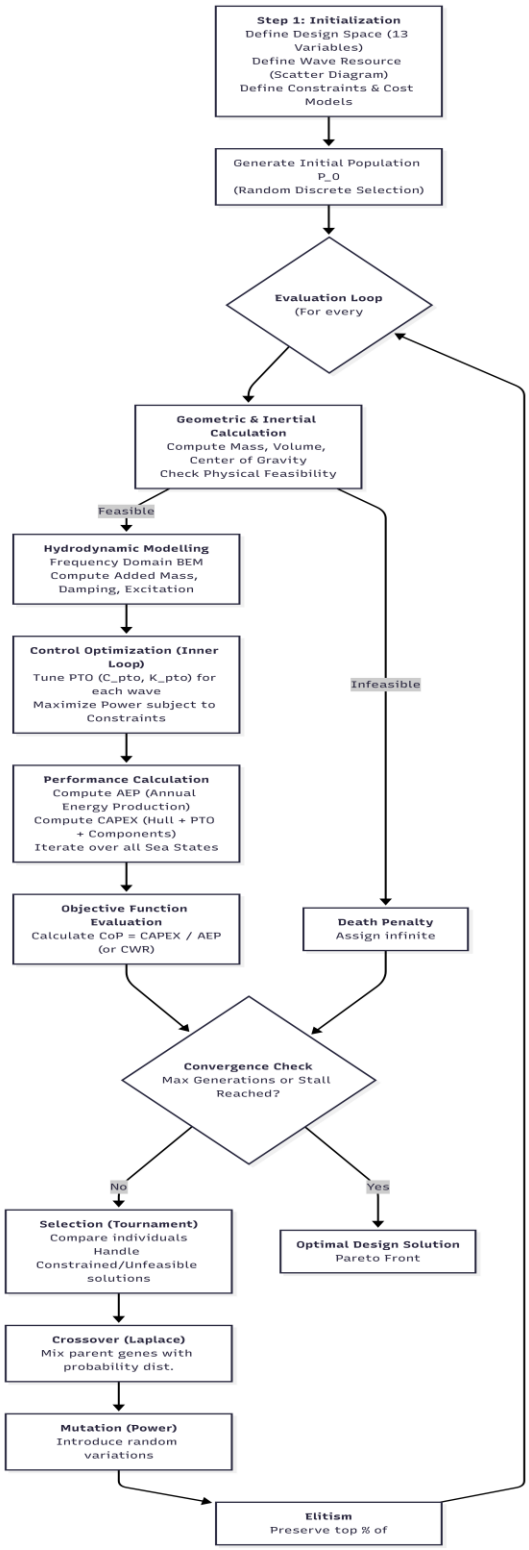


Figure 1. Scheme of the optimization model developed using a genetic algorithm.

Mathematical Formulation

The device is modelled as a multibody system consisting of a floating hull interacting with the incident wave field and an internal pendulum oscillating around a horizontal hinge. To ensure computational efficiency suitable for the iterative nature of the optimization process, a frequency-domain approach is adopted. This method allows for the evaluation of vast design spaces without the prohibitive computational cost associated with time-domain solvers.

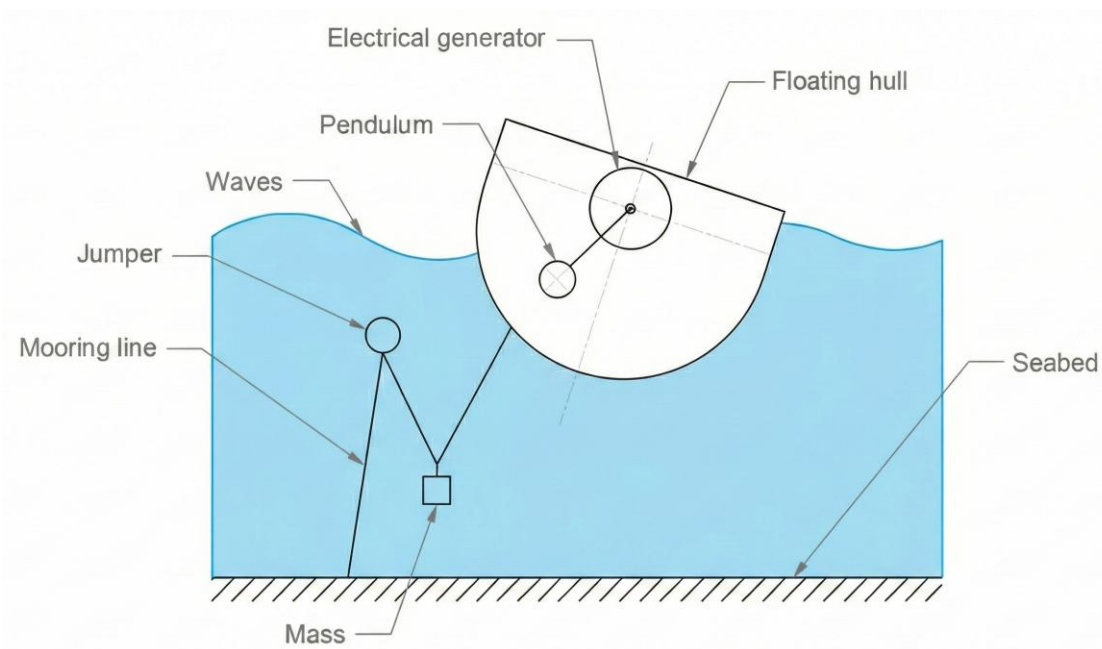


Figure 2. PeWEC concept.

The system dynamics are defined by a generalized coordinates vector q, which includes the hull surge (x), heave (z), pitch (delta), and the relative rotation of the pendulum (epsilon). The relative angle epsilon represents the primary degree of freedom for energy extraction, as the PTO system is coupled to the relative motion between the hull and the pendulum. Under the assumption of linear wave theory and small oscillations, the dynamic equilibrium is solved in the frequency domain.

The equation of motion accounts for the following matrix components:

$$\left[-\omega^2\left(M + A(\omega)\right) + i\omega\left(B_{rad}(\omega) + B_{pto}\right) + \left(K_{hyd} + K_{pto} + K_{moor}\right)\right]\hat{q} = F_{ext}(\omega)$$

- ω: Wave angular frequency
- i: Imaginary unit
- M: Mass matrix (physical mass + inertia)
- A(ω): Hydrodynamic added mass matrix
- B_rad(ω): Hydrodynamic radiation damping matrix
- B_pto: PTO damping matrix
- K_hyd: Hydrostatic stiffness matrix
- K_pto: PTO stiffness matrix
- K_moor: Mooring stiffness matrix
- q̂: Complex amplitude of generalized displacement vector
- F̂_exc(ω): Wave excitation force vector

Mass Matrix (M): Includes the physical mass and moments of inertia of both the hull and the pendulum, as well as inertial coupling terms that drive the energy transfer mechanism.

Hydrodynamic Coefficients: The frequency-dependent added mass matrix A(omega) and radiation damping matrix Brad(omega) are computed using the Boundary Element Method (BEM) solver NEMOH.

Stiffness Terms: Includes the hydrostatic stiffness Khyd (buoyancy restoring forces) and the PTO stiffness Kpto, which is used as a control parameter to tune the resonance frequency of the pendulum.

PTO Damping: The matrix Bpto represents the linear damping torque applied by the generator to extract energy from the relative motion.

Viscous damping effects are neglected in this study to maintain low computational overhead, justified by the dominance of radiation and inertial forces for large-volume structures in the operational frequency range.

Power Production Estimation

The primary objective of the numerical model is to quantify the Annual Energy Production (AEP). This is achieved by first calculating the Response Amplitude Operator (RAO) of the system, which describes the device's response to a unit amplitude wave.

Real ocean conditions are represented by a set of sea states, each defined by a wave spectrum characterized by a significant wave height (Hs) and an energy period (Te).

The average power absorbed by the PTO in a specific sea state is derived by integrating the spectral response. The total AEP is computed by aggregating the weighted power contributions of all representative sea states in the scatter diagram. For the specific case study of Pantelleria Island, up to 128 representative sea states were used. The final calculation includes an efficiency factor of 0.90 to account for electrical and mechanical conversion losses.

Techno-Economic Metrics

To evaluate the true viability of different PeWEC configurations, the optimization model extends beyond simple hydrodynamic efficiency. The ultimate goal is to minimize Capital expenditure over AEP ratio (CoP) [CapEx/MWh], which serves as the objective function for the genetic algorithm. This metric is defined as the ratio of the total capital investment to the energy produced over the device's lifetime, balancing performance with affordability.

Power and Energy Production Estimation

The energy generation potential is quantified through the Annual Energy Production (AEP). This calculation begins by determining the average power absorbed by the device in specific sea states.

The average power absorbed by the Power Take-Off (PTO) system for a given sea state—defined by a significant wave height (H_s) and an energy period (T_e)—is calculated by integrating the spectral response of the device across all wave frequencies 1. The mathematical formulation is:

$$P_{avg}(H_s, T_e) = \int_0^{\infty} \frac{1}{2} C_{pto} \omega^2 |H_\varepsilon(\omega)|^2 S_\eta(\omega) d\omega$$

Where:

- $S_\eta(\omega)$: Is the wave energy spectrum for the specific sea state, representing how wave energy is distributed across different frequencies (ω).
- H_ε(ω): Is the Response Amplitude Operator (RAO) of the relative pendulum motion (ε). It describes the amplitude of the pendulum's oscillation relative to the hull per unit of wave amplitude at each frequency.
- $|H_\varepsilon(\omega)|^2$: Represents the squared magnitude of the response, directly correlating to the energy available in the mechanical oscillation.

- C_pto: Is the scalar damping coefficient of the PTO system. This value represents the torque applied by the electrical generator to extract energy from the pendulum's motion.
- ω^2: A kinematic term relating position to velocity, necessary because power is the product of torque and angular velocity.

Physically, this equation sums up the power extracted from every frequency component of the incoming wave spectrum. It highlights that power production depends not just on the wave height (contained in S_η), but critically on how well the pendulum's resonant response (H_ε) aligns with the wave's peak energy frequencies.

Annual Energy Production (AEP) The total energy generated in a typical year is estimated by aggregating the performance across all sea states that occur at the deployment site. For the Pantelleria Island case study, this involves summing the weighted contributions of representative sea states:

$$AEP = 8760 \cdot \eta_{tot} \cdot \sum_{i=1}^{N_{ss}} P_{el,i}(H_{s,i}, T_{e,i}) \cdot p_i$$

Where: 8760: Represents the total number of hours in a standard year.

- N_ss: Is the total number of sea states considered in the scatter diagram (N_ss = 128 for Pantelleria).
- P_el,i: Is the average electrical power output for the i-th sea state (derived from P_avg).

- p_i: Is the probability of occurrence for that specific sea state (e.g., how often do 2-meter waves with a 6-second period occur?).

- η_tot: Is the global efficiency factor, set to 0.90 (90%). This accounts for losses in the mechanical transmission (gearbox) and electrical conversion (generator and inverter) chain, ensuring the estimate reflects "wire-to-grid" energy rather than just theoretical mechanical absorption.

This probabilistic approach ensures that the device is optimized for the actual local wave climate—prioritizing efficiency in the most frequent sea conditions rather than just the most energetic ones.

Capital Expenditure (CapEx) Estimation The cost of the device is estimated using a parametric model that links physical attributes (mass, torque, dimensions) to component costs. Since the device is in a preliminary design phase, the costs are aggregated into four main categories:

$$\text{CapEx} = C_{hull} + C_{gen}(T_{gen,max}) + C_{bal} + C_{fix}$$

Where:

- C_hull (Structural Cost): This constitutes the largest portion of the investment. It is calculated by determining the total surface area of the hull geometry and multiplying it by a reference steel thickness and material density to find the total steel mass. A specific market unit price for manufactured marine-grade steel is then applied. This incentivizes the optimization algorithm to find shapes that enclose large volumes (for buoyancy) using minimal surface area (to save cost).
- C_gen (PTO Cost): This cost is modelled as a function of the rated generator torque (T_gen,max). The rated torque dictates the physical size and volume of the electrical machine more than its power rating. The cost is derived from a linear regression of commercial component databases, aggregating the price of the entire drivetrain (electric generator, multistage gearbox, and variable frequency drive).
- C_bal (Ballast Cost): Computed based on the mass required for both the active pendulum (typically high-density metal) and the fixed ballast (concrete or sand) needed to adjust the center of gravity and draft.
- C_fix (Fixed Costs): A constant overhead added to account for assembly, sensors, control electronics, and other minor subsystems that do not scale significantly with the device geometry.

Hull Parametrization

A significant advancement in WP1 is the introduction of a B-spline based parametrization for the hull geometry, replacing the previous arc-based approach. This allows for a higher degree of geometric freedom, enabling the optimizer to

explore shapes with varying curvature and volume distribution that are better tuned to the local wave climate.

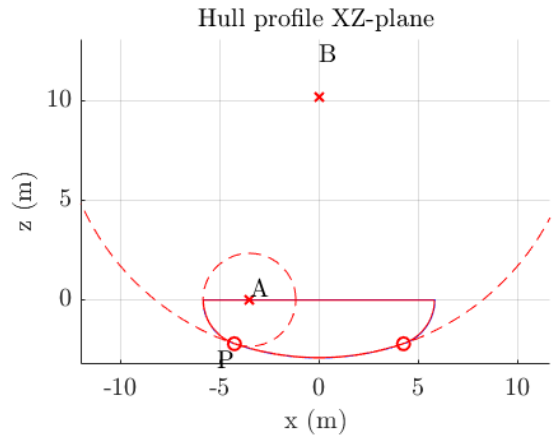


Figure 3. Former hull parametrization using arcs.

The hull profile is generated using a cubic B-spline curve constructed as a weighted sum of basic functions. The profile is defined by six control points and three dimensionless shape parameters (s1, s2, s3). This parametrization reduces the complex problem of shape optimization to three variables while ensuring the hull profile remains symmetric and continuous.

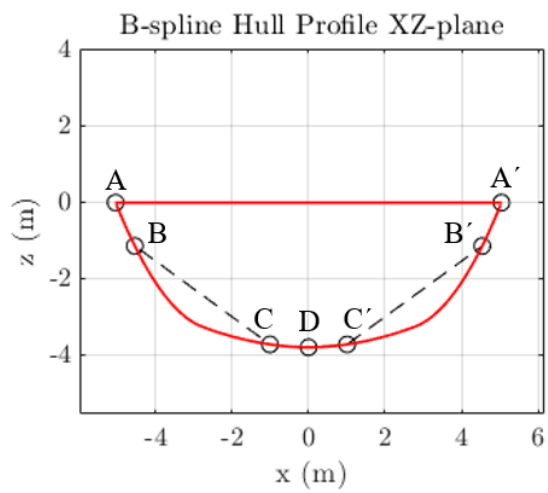


Figure 4. New spline-based parametrization of the PeWEC hull.

To overcome the limitations of simple arc-based shapes, the hull profile is generated using a **cubic B-spline curve**. This method provides a higher degree of geometric freedom, allowing the optimizer to tune the hull's curvature to the specific local wave climate.

The hull profile, denoted as S(u), is constructed as a weighted sum of **control points** (P_i). The influence of each control point on the curve is determined by **basis functions** (B_i), which are piecewise cubic polynomials.

Key Features:

Clamped Knot Vector: Ensures the curve starts exactly at the bow and ends at the stern.

Symmetry: The hull is symmetric fore-and-aft, reducing the complexity of the problem.

3 Shape Parameters: The entire shape is controlled by just three dimensionless variables (s_1, s_2, s_3) along with the total Length (L) and Height (H).

1. The B-Spline Equation

This formula calculates the final curve S(u) by summing the 6 control points multiplied by their basis functions.

$$S(u) = \sum_{i=1}^{6} P_i \cdot B_i(u)$$

2. The Unity Condition

This ensures that the weights always add up to 1, keeping the curve predictable.

$$\sum_{i=1}^{6} B_i(u) = 1 \quad \text{for all } u \in [0, 1]$$

3. Control Point Coordinates

These formulas define exactly where the 6 control points sit in 2D space based on your inputs (L, H, s_1, s_2, s_3).

P1 (Stern/Waterline):

Plaintext

```
P_1 = (-L/2, 0)
```

P2 (Intermediate Stern): Controls the curvature near the water surface.

Plaintext

```
P_2 = (L s_1 - L/2, -H s_2)
```

P3 (Bottom Stern): Defines the start of the flat bottom section.

Plaintext

```
P_3 = (L s_3 - L/2, -H)
```

P4 (Bottom Bow): Defines the end of the flat bottom section.

Plaintext

```
P_4 = (L s_3 + L/2, -H)
```

P5 (Intermediate Bow): Symmetric to P2.

Plaintext

```
P_5 = (L s_1 + L/2, -H s_2)
```

P6 (Bow/Waterline):

Plaintext

```
P_6 = (L/2, 0)
```

Physical Meaning of the Parameters

s_1 (Horizontal Curvature): Controls how "full" or "slender" the hull is near the waterline. Increasing s_1 moves the intermediate points (P_2, P_5) outward, making the hull boxier.

s_2 (Vertical Curvature): Controls the vertical position of the curvature. If s_2 is high, the curve stays shallow; if low, it cuts deeper towards the keel.

s_3 (Flat Bottom Width): Controls the width of the flat bottom section (P_3 to P_4). A larger s_3 creates a barge-like flat bottom, while a smaller s_3 creates a V-shape or rounded keel.

Strict validity constraints are applied to ensure manufacturability and hydrostatic stability:

Monotonicity Constraint: Ensures the hull profile curves consistently without creating pockets or re-entrant shapes. For the submerged profile, the slope must never be positive, preventing geometries that would be difficult to manufacture via LSAM or hydrodynamically unstable.

Geometric Consistency: A height constraint ensures that the intermediate control points do not drop below the keel or rise too high, guaranteeing a convex or semi-convex shape appropriate for a pitching body and securing sufficient internal volume for the pendulum swing.

Preliminary results

Optimization Settings The minimization of the CoP is driven by a Genetic Algorithm (GA) employing a tournament-based NSGA-II approach. This metaheuristic engine is well-suited for high-dimensionality problems where the objective function is non-differentiable.

The optimization was configured with the following specific settings:

- Population Size: 70 individuals per generation.
- Maximum Generations: 150 iterations.
- Design Space: The algorithm explores 16 variables, including hull dimensions (Length 10-30m, Width 5-25m), inertial mass distribution, and PTO settings.

The optimization runs were computationally intensive, requiring approximately 96 hours per full run on a high-performance workstation. The model incorporates a rigorous constraint handling mechanism where configurations violating physical or geometric rules are assigned a "death penalty," effectively removing infeasible solutions from the gene pool. Notably, the spline-based optimization initially presented a high mortality rate of 99% due to the random generation of invalid shapes, but this stabilized to between 30% and 50% as the algorithm identified feasible regions of the design space.

The comparative analysis of the two optimal individuals demonstrates that the spline-based geometry outperforms the benchmark arc-based design across all key techno-economic metrics.

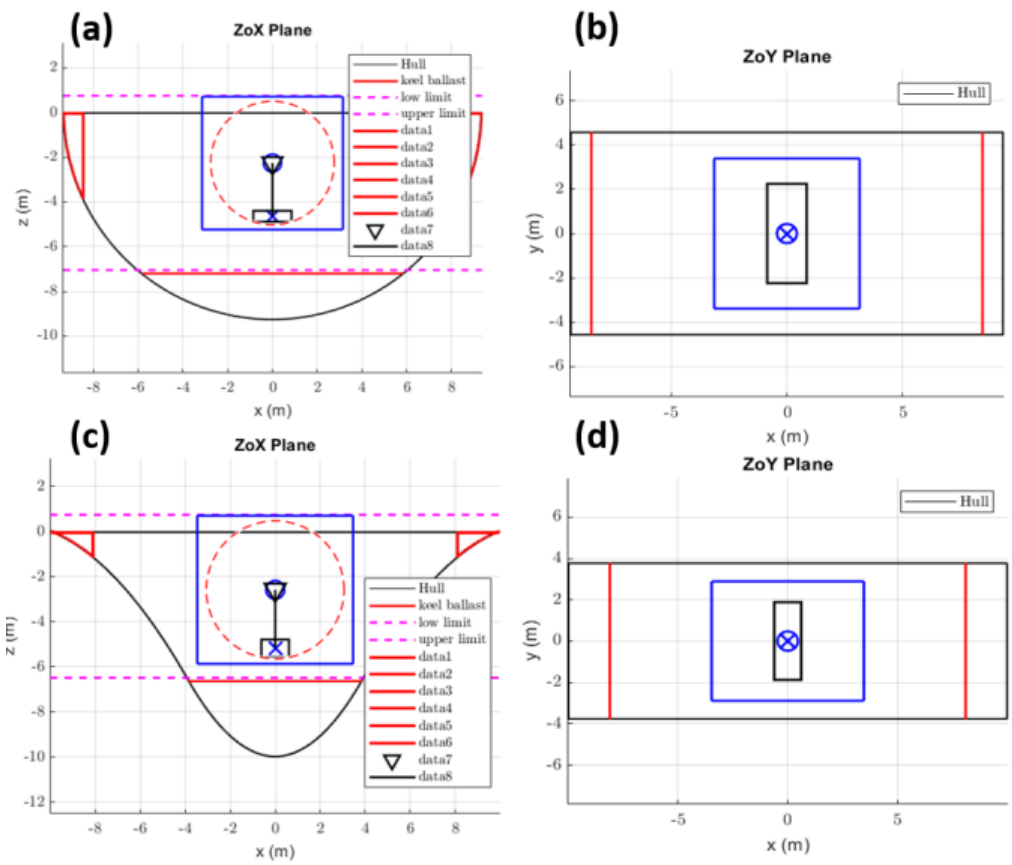


Figure 5. Example of optimal hulls shapes: (a) side view with design elements and bounds of an arc-based hull; (b) relative top view of the arc-based hull; (c) side view with design elements and bounds of and (d) top view of the spline-based hull.

Regarding the primary objective, the Cost of Energy (CoP), the spline-based individual achieves a lower (more efficient) ratio by accessing a design space capable of higher energy yields—reaching an Annual Energy Production (AEP) of up to 80 MWh compared to the 60 MWh limit of the arc-based design. This superior economic performance is driven by hydrodynamic gains: the spline-based hull exhibits a significantly higher Annual Capture Length Ratio (aCLR), approximately 0.2 higher than its arc-based equivalent. This indicates that the spline shape utilizes the device's length more effectively to couple with wave forces. While the Annual Capture Width Ratio (aCWR) for the arc-based design is concentrated in a narrow performance band (45-55 MWh), the spline geometry maintains high efficiency across a broader range . These improvements are physically attributed to the spline parametrization allowing for a more slender hull (7.56 m width vs 9.19 m) with

optimized curvature, which reduces pitch radiation damping and enhances the resonant response.

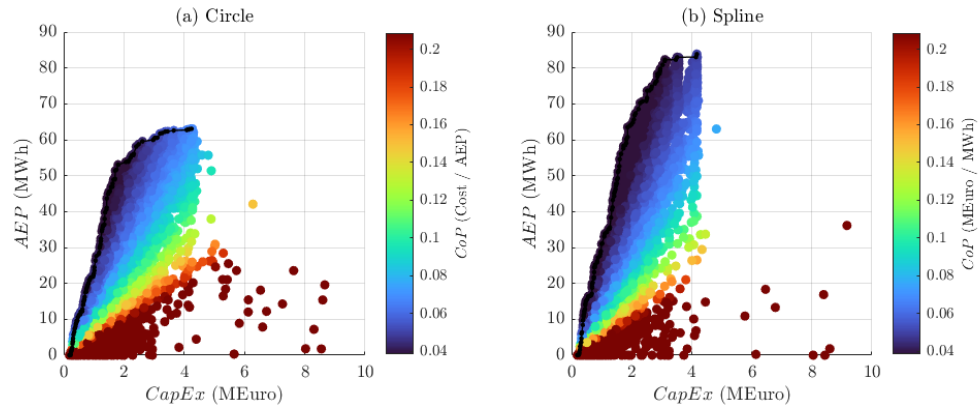


Figure 6. CoP results for the two 5th best individuals: (a) arc-based hull; (b) spline-based hull

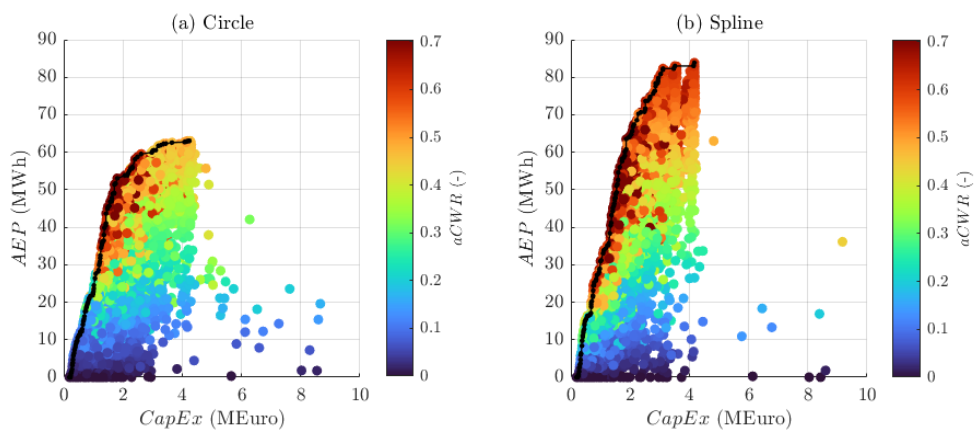


Figure 7. Annual Capture Width Ratio (aCWR) results for the two 5th best individuals: (a) arc-based hull; (b) spline-based hull

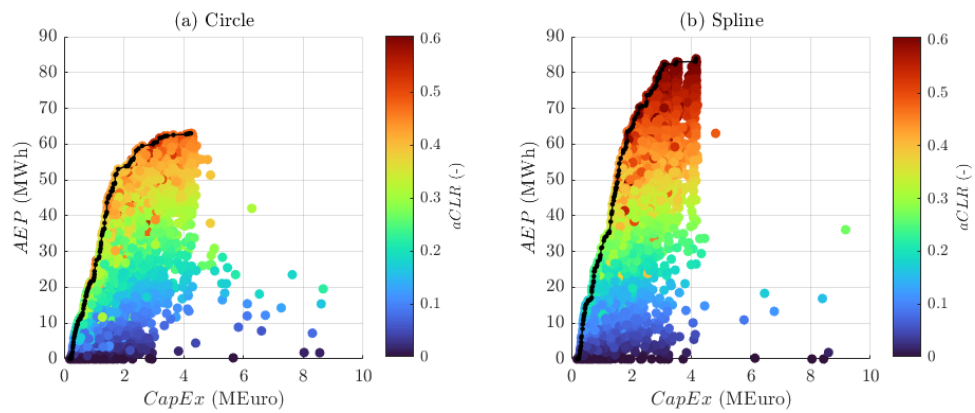


Figure 8.Annual Capture Length Ratio (aCLR) results for the two 5th best individuals: (a) arc-based hull; (b) spline-based hull

The hydrodynamic analysis reveals distinct differences in the dynamic response characteristics of the two optimal geometries. The spline-based hull exhibits notably lower pitch added mass (A) and reduced radiation damping (B) compared to the arc-based design[1]. This reduction in radiation damping is particularly critical, as it implies that the hull dissipates less energy back into the water through wave generation during oscillation. Consequently, this reduced damping allows the spline-based design to achieve a significantly higher and narrower peak in its Response Amplitude Operator (RAO). Physically, this behaviour indicates that the spline-based geometry functions as a higher-quality resonator, capable of tuning itself more effectively to the incoming wave spectrum and amplifying the pitch motion required for energy extraction.

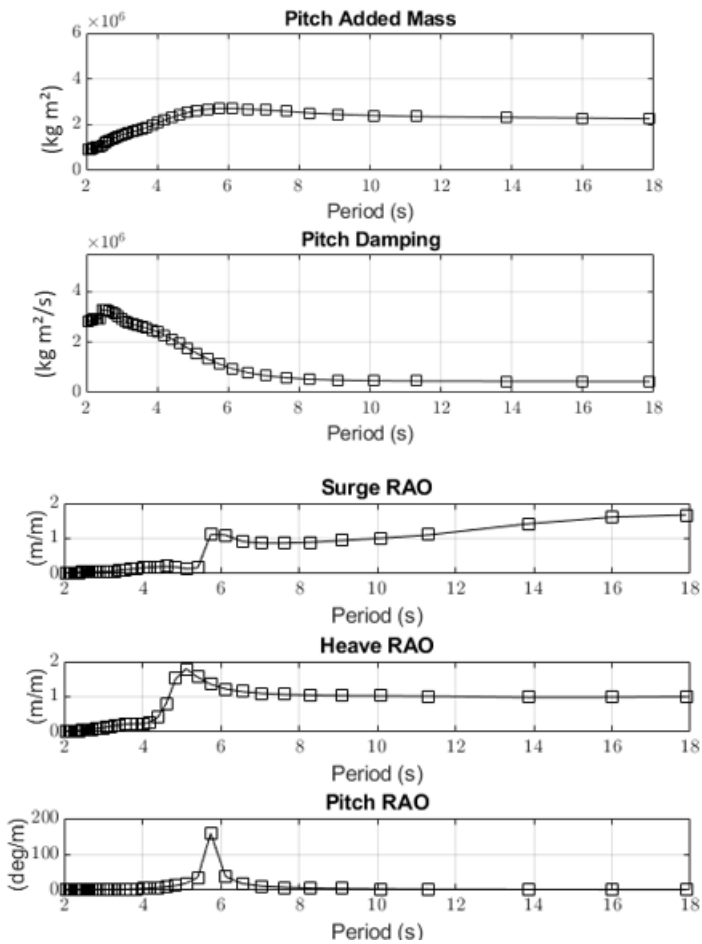


Figure 9. Pitch added mass, radiation damping and RAOs for the spline-based hull.

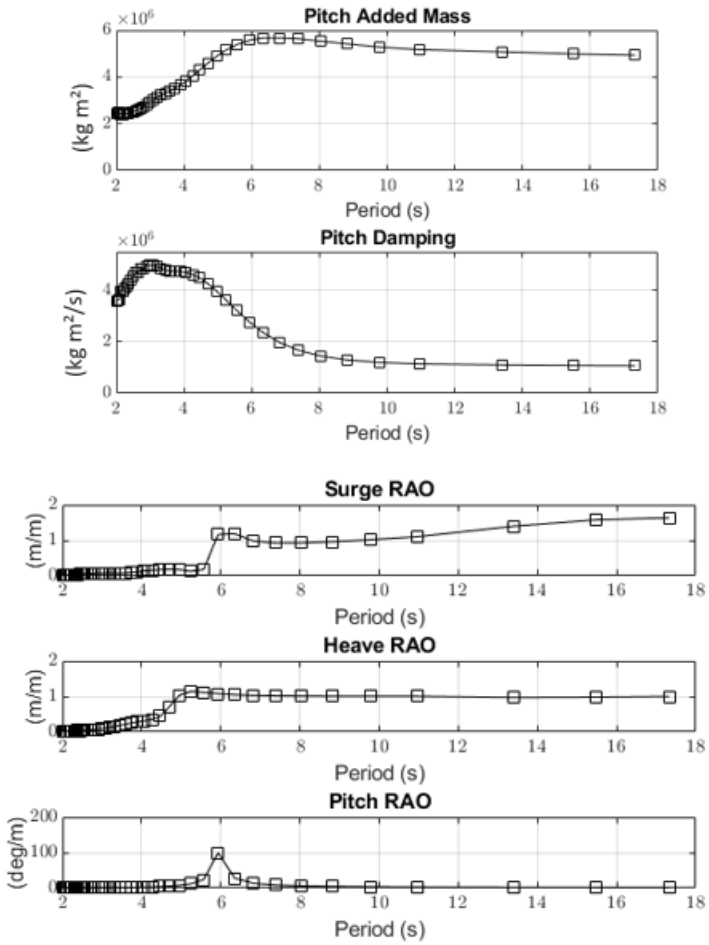
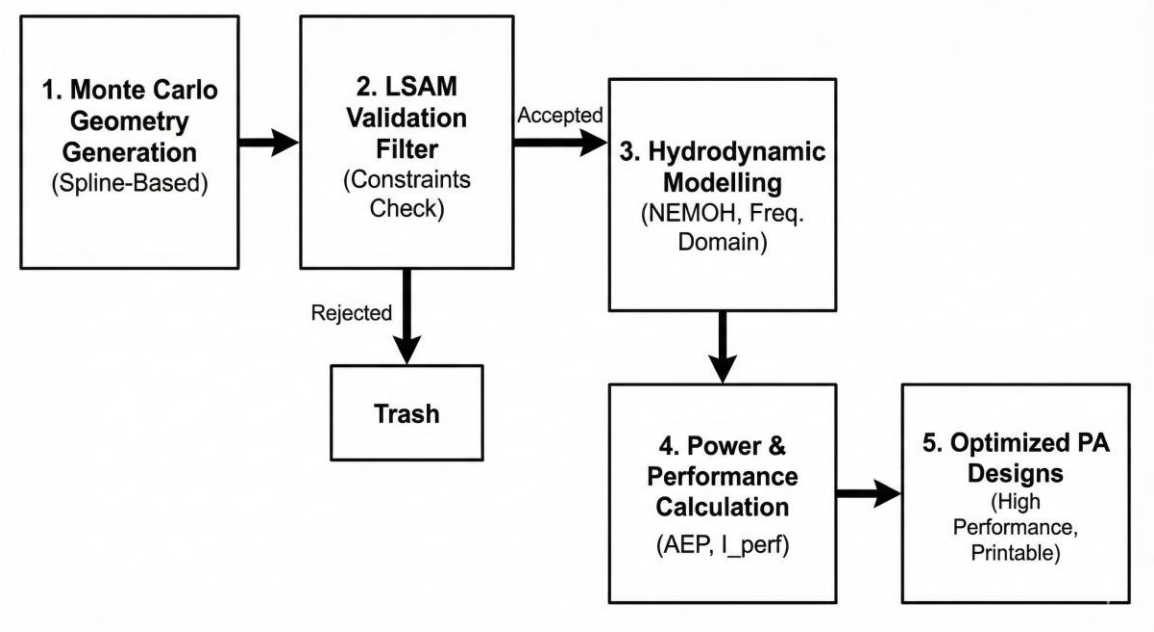


Figure 10. Pitch added mass, radiation damping and RAOs for the arc-based hull.

9. Numerical modelling and development of the experimental PA

This section outlines the numerical framework developed for the optimization of the Point Absorber (PA) case study. Unlike the evolutionary Genetic Algorithm approach used for the PeWEC, the PA development employs a Monte Carlo simulation framework. This method relies on the stochastic generation of a vast number of geometries, which are rigorously filtered and evaluated to explore the design space comprehensively and identify high-performance candidates suitable for Large-Scale Additive Manufacturing (LSAM).

The numerical optimization of the Point Absorber (PA) follows a fully automated workflow designed to identify high-performance geometries compatible with LSAM. The process initiates with a Monte Carlo algorithm that stochastically generates a diverse population of hull profiles using cubic B-splines. These candidates are immediately subjected to a rigorous LSAM validation filter, which discards any shapes violating manufacturability constraints such as monotonicity or excessive overhangs. Accepted geometries proceed to hydrodynamic modelling, where the NEMOH BEM solver computes the necessary coefficients (added mass, damping, excitation) in the frequency domain. Subsequently, the energy performance is evaluated by integrating power output across the specific wave climate of the deployment site to determine the Annual Energy Production (AEP), while simultaneously calculating a specialized Index of Performance (I_{perf}) that normalizes energy yield against the wet surface area. This comprehensive pipeline ultimately outputs a ranked selection of PA designs that maximize techno-economic efficiency while ensuring physical printability.



Automated Simulation Framework The computational workflow is fully automated to handle large datasets of potential designs. The process follows a sequential structure:

Geometry Generation: A dedicated script generates a large population of random hull profiles defined by B-splines (higher order splines).

LSAM Validation: Each generated geometry undergoes an automated screening process. Profiles that do not comply with manufacturing constraints (e.g., those with non-monotonic curvature, excessive overhangs, or self-intersections) are immediately classified as "rejected" and removed from numerical testing.

Hydrodynamic Evaluation: Only the "accepted" geometries—those strictly compliant with LSAM requirements—are passed to the hydrodynamic solver for performance assessment.

Equation of motion

The dynamic behaviour of the Point Absorber is governed by the following linear equation of motion, which balances the inertial, damping, and stiffness forces against the wave excitation force.

$$\left[-\omega^2\left(M + A(\omega)\right) + i\omega\left(B_{rad}(\omega) + B_{pto}\right) + \left(K_{hyd} + K_{pto} + K_{moor}\right)\right]\hat{q} = \hat{F}_{exc}(\omega)$$

Explanation of Terms:

ω: Wave angular frequency [rad/s].

i: Imaginary unit.

q̂: Complex amplitude vector of the device's oscillation (e.g., heave, surge, pitch).

F̂_exc(ω): Vector of the wave excitation force acting on the hull.

Inertial Terms (Acceleration)

M: Physical Mass Matrix. This represents the actual mass and inertia of the floating hull.

A(ω): Hydrodynamic Added Mass Matrix. This represents the inertia of the water surrounding the hull that must be accelerated along with the device.

Damping Terms (Velocity)

B_rad(ω): Hydrodynamic Radiation Damping Matrix. This accounts for the energy dissipated by the hull creating waves as it oscillates.

B_pto: PTO Damping Matrix. This represents the useful damping applied by the Power Take-Off system (the generator) to extract energy from the motion.

Stiffness Terms (Position)

K_hyd: Hydrostatic Stiffness Matrix. This represents the buoyancy restoring force that tries to return the hull to its equilibrium position.

K_pto: PTO Stiffness Matrix. An optional control parameter used to tune the device's resonance frequency to match the incoming waves.

K_moor: Mooring Stiffness Matrix. Represents the restoring forces provided by the anchor lines (often neglected in preliminary power optimization if they are compliant).

Geometry Definition and LSAM Constraints

The hull geometry of the PA is defined as an axisymmetric body of revolution. The profile of the floater is parameterized using cubic B-splines, controlled by a set of variable control points and shape parameters. This allows for the generation of complex, organic shapes that differ significantly from standard cylindrical or conical buoys.

To ensure the final designs are printable, the code implements strict geometric checks during the generation phase. Key parameters such as the start and end points of the spline (defining the draft and radius) and the derivative at the keel are constrained. For the specific case study analyzed, the reference dimensions were set to a radius of 0.5 m and a maximum draft of 0.75 m, consistent with a target 1:10 scale for physical model testing.

Hydrodynamic Modelling

The physical behavior of the device is simulated in the frequency domain to maintain computational efficiency across the hundreds of iterations (e.g., >450 valid individuals per run). The hydrodynamic coefficients—added mass A(ω), radiation damping B(ω), and excitation forces F_exc(ω)—are computed for each individual geometry using the Boundary Element Method (BEM) solver Nemoh.

The automation script systematically generates the mesh for the specific spline profile and performs all calculations:

- Calculates hydrostatic properties, including **Displacement Volume** and **Wet Surface Area** (S_wet).
- Solves the diffraction and radiation problems for a frequency range of **0.3 to 6.0 rad/s**.
- Identifies the **Natural Period** (T_n) of the device by finding the frequency where the total reactance (hydrostatic stiffness minus inertial terms) approaches zero.
- It calculates the **RAOs**

AEP and Cost Function computations

Successively the model performs power, energy and cost functions calculations.

Power Production and Optimization Metrics - The energy performance is evaluated using a spectral approach. The Annual Energy Production (AEP) is calculated based on the wave climate data from the NOEL laboratory site (Reggio Calabria, Italy),

represented by a detailed occurrence matrix of Significant Wave Heights (Hs) and Peak Periods (Tp).

For every sea state, the model employs a "Lookup Method" to determine the optimal Power Take-Off (PTO) damping coefficient (C_pto). This damping is tuned to maximize power absorption, ideally matching the radiation damping at the resonance frequency or the specific peak frequency of the wave spectrum. The power output is then integrated using a JONSWAP spectrum to account for real sea state energy distribution.

The ultimate goal of this Monte Carlo analysis is not just to maximize gross energy, but to maximize efficiency relative to material usage. Therefore, the code identifies the best-performing individuals based on a specific Index of Performance (I_perf):

I_perf = AEP / Wet Surface Area [MWh/m^2]

This metric serves as a proxy for the Techno-Economic efficiency, rewarding designs that generate high power while minimizing the surface area (and thus the LSAM printing material and time) required for the hull.

Developed Numerical Scripts

The simulation framework is built upon a modular set of MATLAB scripts, each handling a specific stage of the workflow. The interaction between these scripts is orchestrated by the main execution file **mainWorkFlow.m**.

mainWorkFlow.m: The central engine of the simulation. It initializes the environment, executes the Monte Carlo generation, loops through every valid geometry to run the hydrodynamic solver (NemohRunner), calculates the natural period (T_n), determines the optimal damping (C_pto) for resonance, runs power, energy and cost metrices calculations and finally aggregates all results into a summary table.

initialParameters.m: This script defines the global physical constants and constraints used throughout the study. It sets the water density (ρ = 1025 kg/m^3), gravity (g), and the specific geometric boundaries for the Point Absorber, such as the maximum draft (0.75 m) and radius (0.5 m). It also defines the frequency vector (ω) for the BEM simulations.

main_spline_mc.m: Responsible for the stochastic generation of hull profiles. It produces a large population of random B-spline geometries and applies the LSAM

filters. Geometries that fail checks for monotonicity or printability are discarded, while valid ones are saved in the "accepted" structure for further analysis.

AxisymSpline.m: A wrapper script called within the main loop. It takes the parameters of a specific valid spline and generates the discretized mesh files required by NEMOH. It outputs the 'Hydrostatics.dat' and 'Mesh.tec' files used to calculate buoyancy and surface area.

calcMeshArea.m: A utility function that parses the mesh file to accurately calculate the total **Wet Surface Area** (S_wet). It sums the area of the triangular and quadrilateral panels defining the hull, which is a critical denominator for the Index of Performance.

NemohRunner.m: The interface between MATLAB and the NEMOH BEM solver. It executes the pre-processor, solver, and post-processor to compute the hydrodynamic coefficients (Added Mass A, Radiation Damping B, and Excitation Force Fe) for the specific mesh generated in the previous step.

AEP_calc.m: This script performs the energy yield assessment. It loads the scatter diagram (occurrence matrix) for the deployment site and calculates the power output for each sea state using a JONSWAP spectrum. It integrates the spectral response to determine the total **Annual Energy Production** (AEP) in MWh/year.

initializeResStruct_0.m: Sets up the data structure for storing results. It defines the table headers and units for all output variables (e.g., Displacement, Draft, AEP, I_perf), ensuring that data from thousands of iterations is organized systematically for post-processing.

Case study results

Explanation of the Point Absorber (PA) Optimisation Process

This section of the WEC-Pro WP1 report details the numerical framework and experimental development for the Point Absorber (PA) wave energy converter. Unlike the Genetic Algorithm approach used for the PeWEC device, the PA employs a stochastic Monte Carlo simulation framework.

Here is a detailed breakdown of the parameters and methodology adopted:

Target Dimensions: The design space is constrained to specific reference dimensions to prepare for physical 1:10 scale model testing in Work Package 3. The target diameter is 1 m (a radius of 0.5 m) with a maximum draft of 0.75 m.

Power Take-Off (Cpto): The system assumes an ideal, optimal Power Take-Off damping coefficient (**Cpto**) based on theoretical formulas. This means the damping is tuned to maximise power absorption, considering the radiation damping at the specific resonance frequency of the incoming waves.

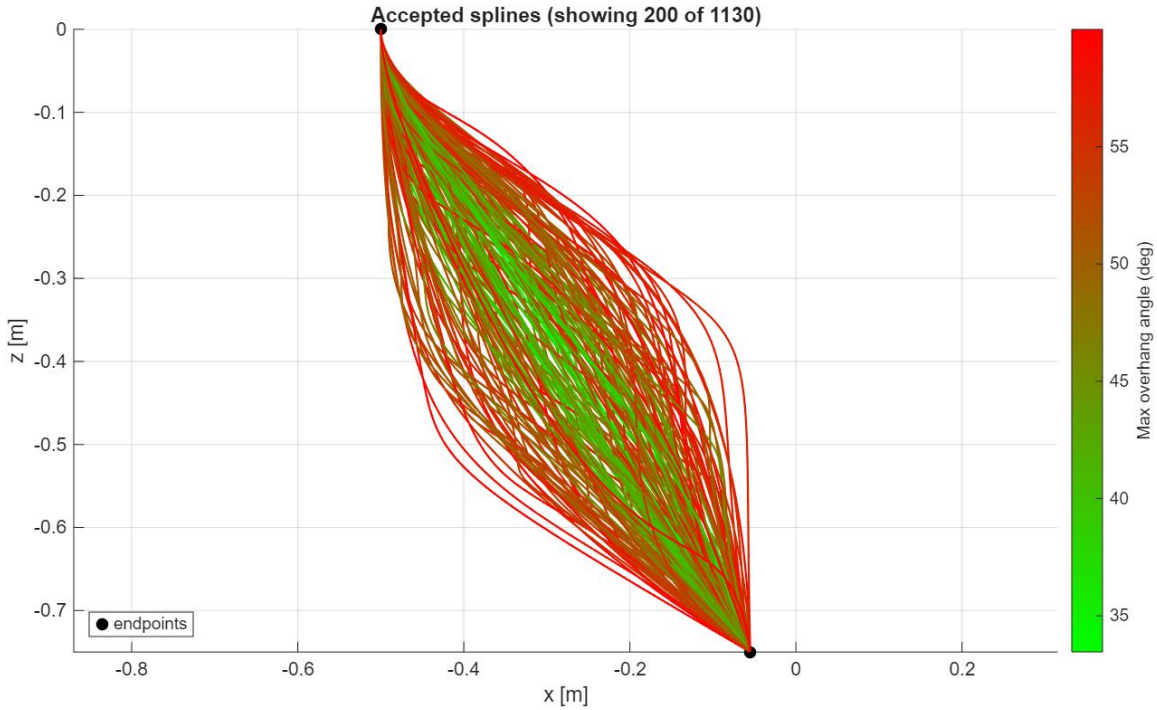
Wave Climate Localisation: The hull shapes are not universally optimised; rather, they are specifically tailored to the real-world wave climate data of the NOEL laboratory deployment site in Reggio Calabria.

The Monte Carlo & LSAM Filter Workflow: The automated script generates thousands of random hull geometries using cubic B-splines to explore complex, organic shapes. Before any hydrodynamic testing occurs, these candidates must pass a rigorous Large-Scale Additive Manufacturing (LSAM) validation filter.

Monotonicity: The curve of the hull must be continuous without self-intersections or re-entrant pockets in the z and x axes (Monotone z and x).

Overhang Limits: To ensure the shape can physically be 3D printed without collapsing, the geometry cannot have an overhang angle exceeding 60.0°.

Hydrodynamic Analysis: Out of 50,000 generated splines, only 1,130 shapes successfully passed the LSAM printability constraints. Only these "accepted" designs proceed to the computationally heavy hydrodynamic evaluation using the NEMOH solver, which assesses their Annual Energy Production (AEP) against the material required to print them.



Simulation Metric	Initial Test Run	Extended Test Run
Total Candidates Tested	20,000	50,000
Accepted Candidates	448	1,130
Acceptance Criteria	Angle <= 60.0 degrees, Monotone z & x	Angle <= 60.0 degrees, Monotone z & x
Best Maximum Angle	32.902 degrees	32.902 degrees
Curvature Cost	34.7	34.7
Final Selection: Overhang Limit	Limit 60.0 degrees -- OK	Limit 60.0 degrees -- OK

Simulation Metric	Initial Test Run	Extended Test Run
Final Selection: Monotone z Check	YES (dz/dt <= 1.0e-12)	YES (dz/dt <= 1.0e-12)
Final Selection: Monotone x Check	YES (dx/dt >= -1.0e-12)	YES (dx/dt >= -1.0e-12)

10. Deliverables and Resources

The following are the main output of this WP:

- The present report covering the work carried out in WP1.
- Open demo script for the Point Absorber Optimization. Accessible on Zotero.
- EWTEC 2025 Conference paper on the study: Multi-Objective Development of a Point Absorber Wave Energy Converter Using Dynamic Modelling. Accessible at the following link: https://www.researchgate.net/publication/395514057_Multi-Objective_Development_of_a_Point_Absorber_Wave_Energy_Converter_Using_Dynamic_Modelling
- OMAE 2026 Conference paper on the study: Optimization of the Pendulum Wave Energy Converter with Spline-Based Hull Shape.
- Additional information can be found at project linked website: waveng.eu